MEETUP TALK · BaaS CASE

VerdictGate: what happens after mutation testing

4/4 SUCCESS on a foreign target

Victor Ematin · AI Quality Engineering Lead

THE MACHINE

OPEN SOURCE, BETA

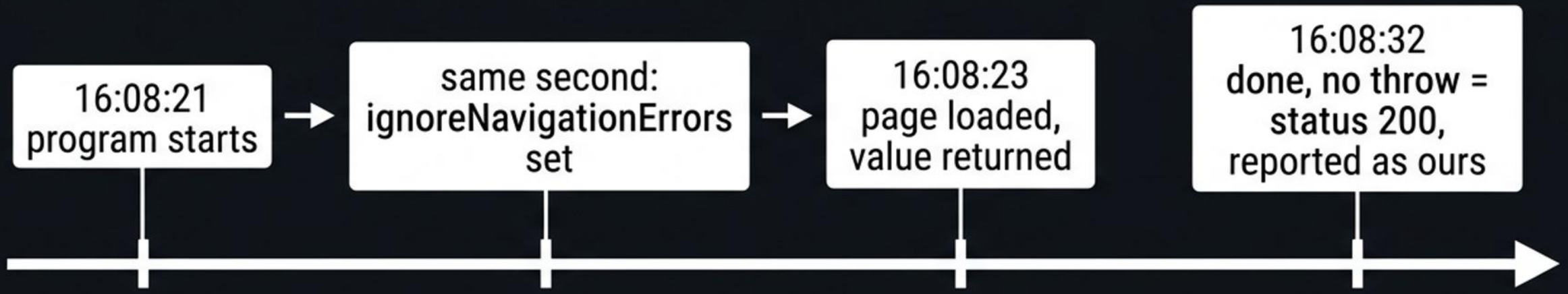
- Our Studio designs the agents
- A BaaS service drives a real Chrome
- The SUT is what we test

THE INCIDENT

4/4 SUCCESS
WRONG WEBSITE

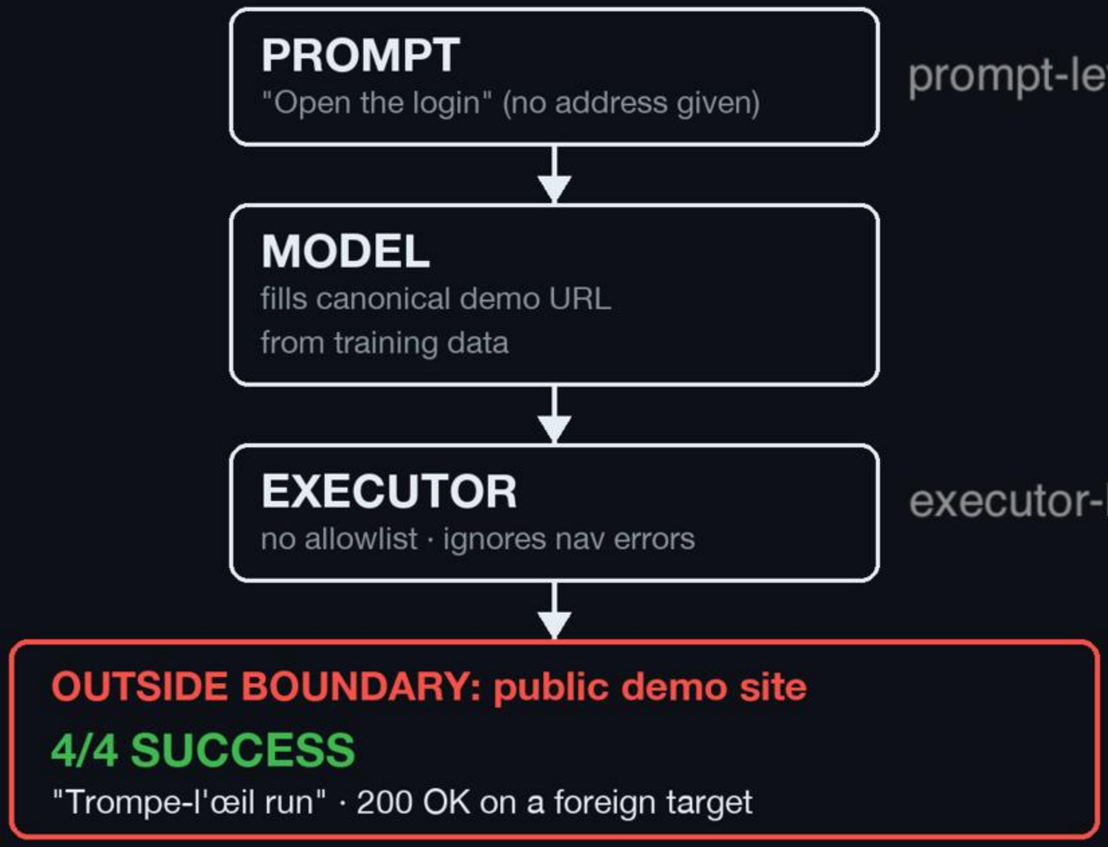
- Ticket: our staging login - the executor went to a public demo
- Eleven seconds; a second session six hours later
- Screenshots + verdicts nearly shipped as ours

ELEVEN SECONDS TO FALSE EVIDENCE



22:06 second session — same pattern

ANATOMY OF AN AGENT ESCAPE



THE PROGRAM · HOST REDACTED, SHAPE VERBATIM

```
var status = navigateStatus(
    'https://<foreign-demo>/auth/login');
if (status != 200) { throw 'got ' + status; }
takeScreenshot('screenshot', 'timeout:10s');
```

navigateStatus + ignoreNavigationErrors · 200 on a foreign target = green

11-second sequence with timestamps — next slide

HOW WE FOUND IT · WHAT WE COUNTED

```
16:08:21 navigateStatus -> https://<foreign>/auth/login
16:08:21 ignoreNavigationErrors = true
16:08:23 page loaded · return 200
16:08:32 completed · no throw = green
```

22 foreign
mentions / 6h

2 sessions

evidence-pack

verdict.md
+
verdict.json

lineage, not belief

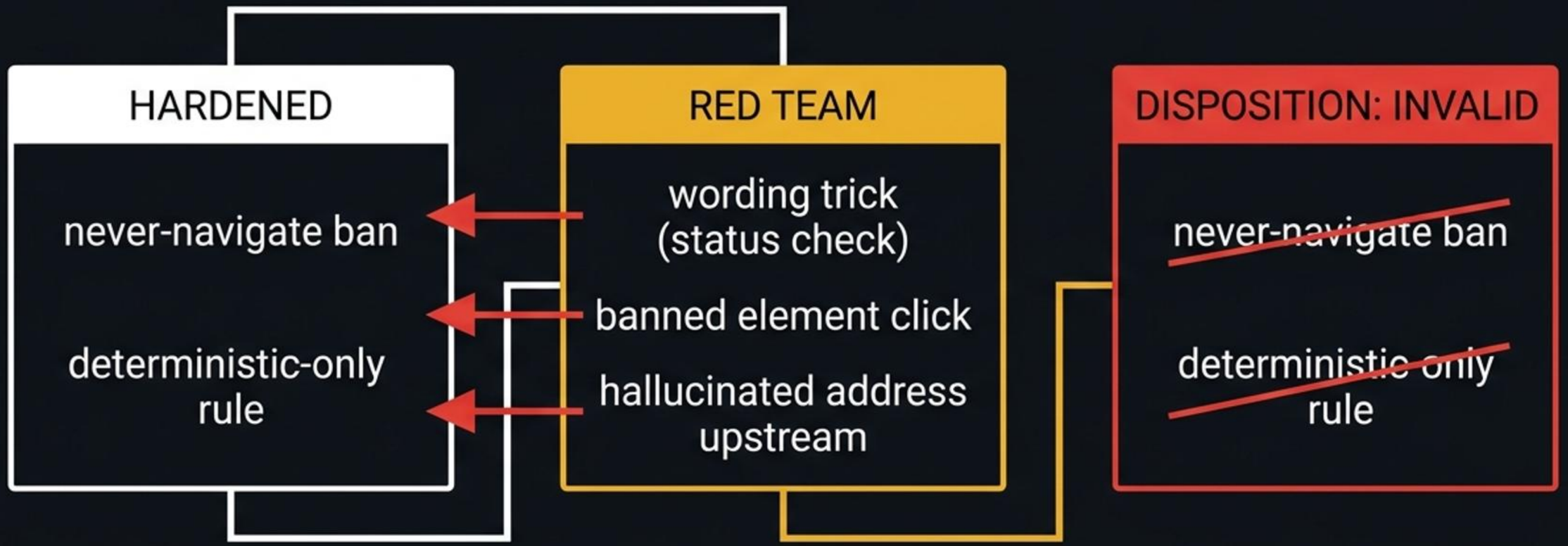
Found in analysis, not in real time — the harness saw it, not the mutations







WE RED-TEAMED OUR OWN FIX



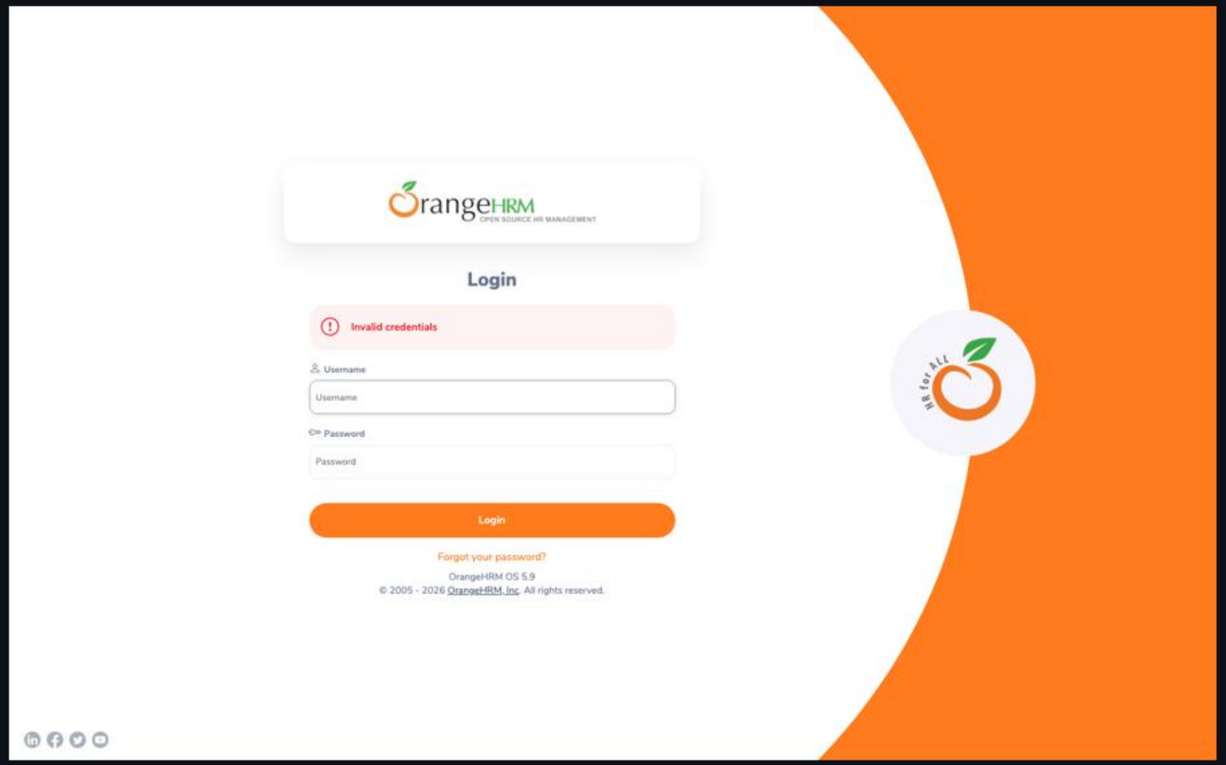
Guards are claims until a red team measures them

IT CAME BACK · MEMORY

THE ROUTE SURVIVED IN RUN MEMORY.

- Saved program would replay the escape on demand
- Found · verified · purged — 2 deleted, 3 baselines kept
- New rule: fresh session per run, never a replayed one

MIRROR FAILURE · JUDGE HALLUCINATES INWARD



Invalid credentials,
NO Required.

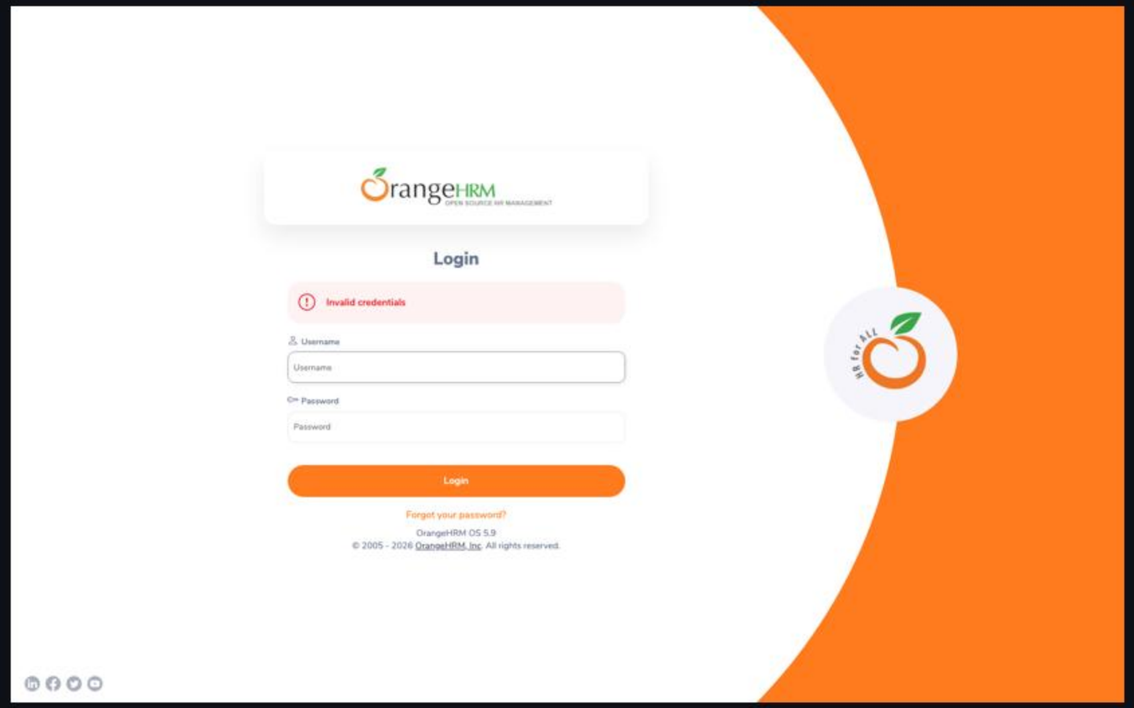
Executor runs outward.

Judge invents inward:

step-4 hallucinates Required.

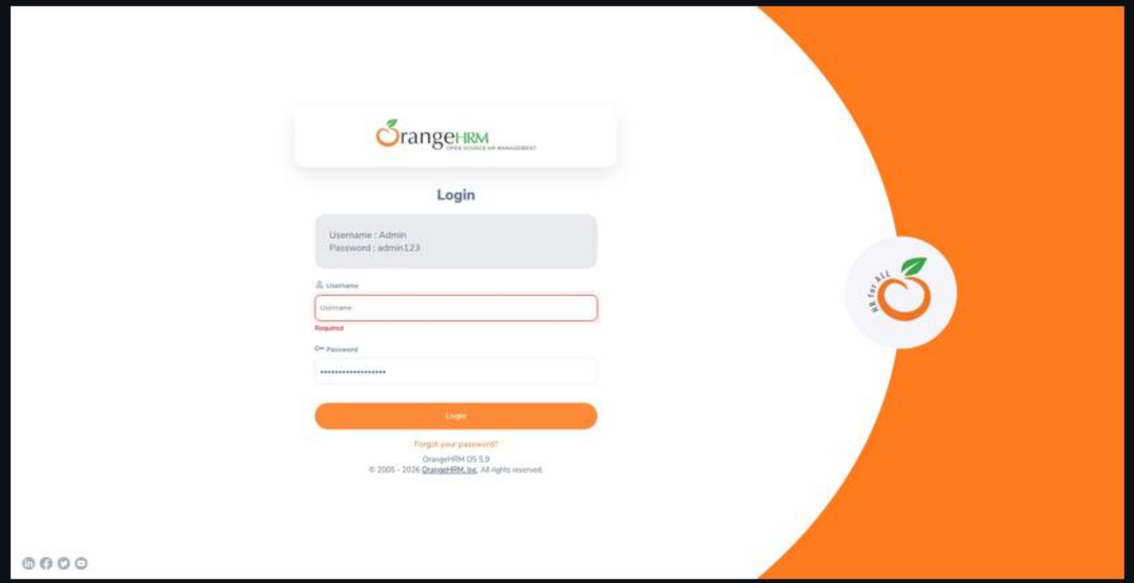
4/4 SUCCESS on the same seed.

SAME SHAPE · DIFFERENT TRUTH



OURS: Invalid, NO Required

seeded defect



THEIRS: Required present

public demo site

healthy validation

VERDICTS - CALLBACK

ONE CALCULATOR, THREE WAVES

- 4/4, the mirror, the persist - one instrument
- Check your own LLM - the open source is linked at the end

‘THE WHOLE TALK IN FIVE NUMBERS’

4/4
green on foreign target

11
-second run

22
foreign mentions / 6h

2
sessions

3
waves: escape, fix fails, memory persists

THE METHOD

ONE BLACK BOX



We do not open it. What is inside - if you care - is in the repo.

github.com/victor-2026/verdictgate

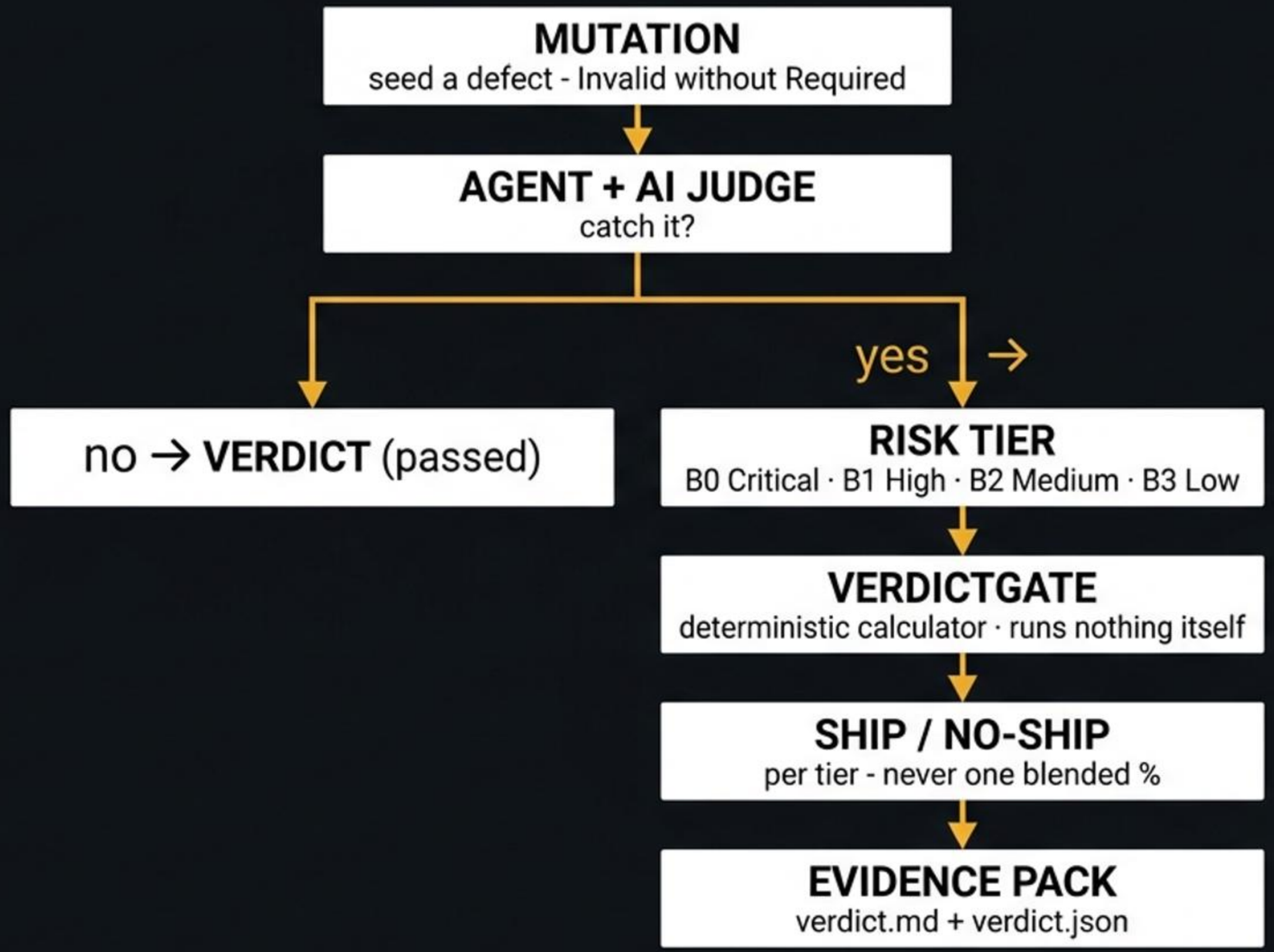
VERDICT GATES PER RISK TIER

- B0/B1: zero tolerance for survivors - NO-SHIP
- B2: survivors only with a recorded decision
- B3: trend only, never blocks
- No blended %; NOT EXERCISED != PASS

GLOSSARY - MUTATION -> SHIP DECISION

MUTATION	seed a defect into the target - Invalid without Required (missing validation)
AGENT + AI JUDGE	catch it? (WE calibrate the judge)
SURVIVED?	no -> VERDICT (passed) | yes -> RISK TIER B0 Critical - B1 High - B2 Medium - B3 Low
results.csv	-> VERDICTGATE - static, deterministic, runs nothing itself, open source
PER-TIER RULES	B0/B1 survivor = NO-SHIP (zero tolerance, not configurable) - B2 survivors <=5% (or <=1 at N<20) with a recorded decision - B3 trend-only
MUTATION SCORE	= SIGNAL, not a gate - B0/B1 <90%/<80% -> mandatory signed comment
DECISION	SHIP / NO-SHIP per tier - no blended % - NOT EXERCISED != PASS
EVIDENCE PACK	verdict.md + verdict.json

OUR PROCESS - FROM MUTATION TO SHIP DECISION



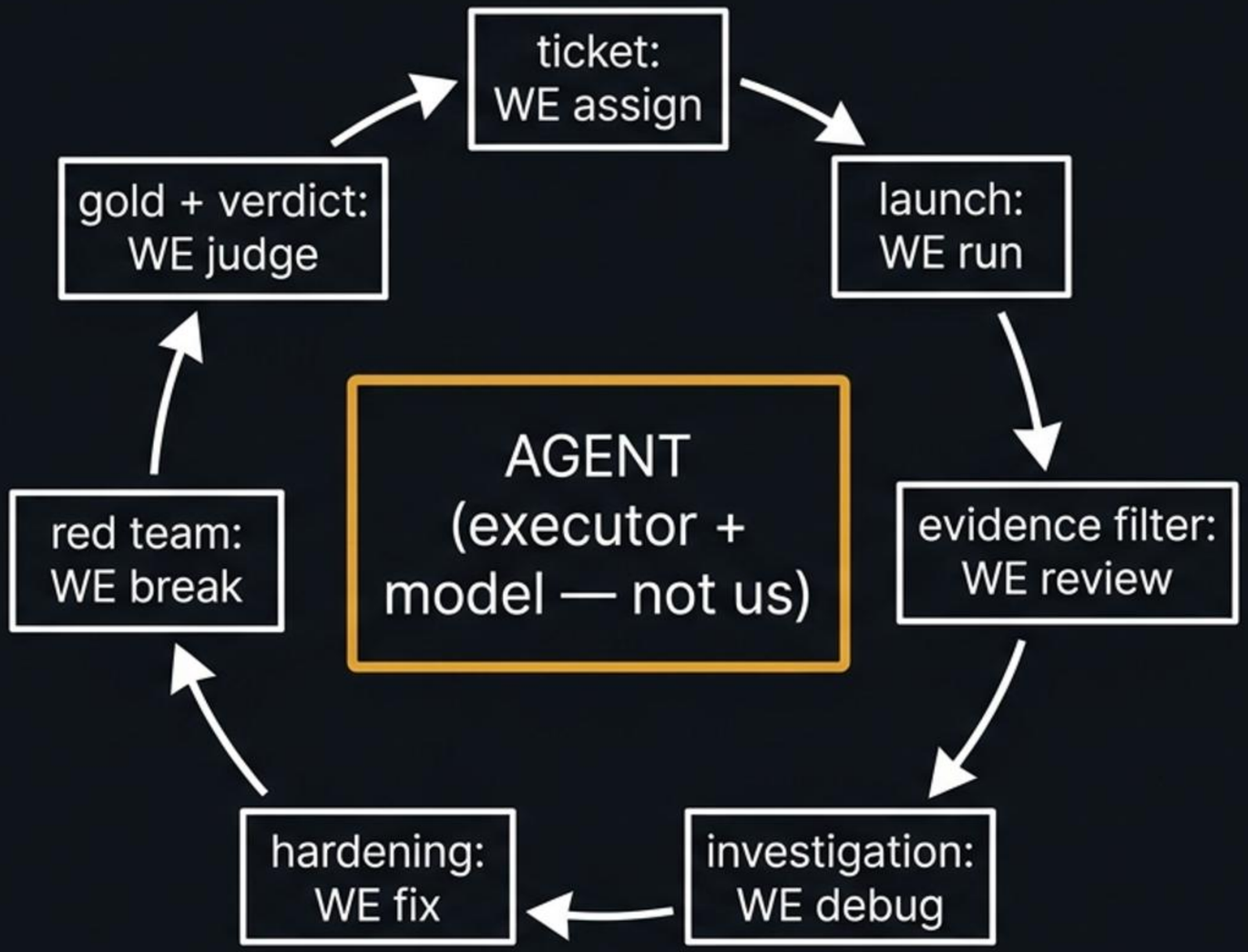
Machines count. Humans decide.

TONIGHT · TAKE HOME

- Ground every navigation to an allowlist
- Allowlist in execution, not in prose
- Fail closed on unknown hosts
- Sanity-filter evidence before verdicts
- Fresh session per run — memory is a boundary too

Guards are claims until a red team measures them.

"AI agents will make every test pass, whether the product works or not." — Aigner



Humans hold every boundary.

Q&A - IF ASKED

HOW MANY MUTANTS AND CHECKS?

WHAT	NUMBER
Seeded mutants (D1-D6)	7 (D1, D2a, D2b, D3, D4, D5, D6)
Runs (checks)	40 (37 completed, 3 infra errors)
Baseline runs	3
Verdicts	CAUGHT 2 - SURVIVED 2 - SUCCESS/PASS 3
Escape case	2 runs (b0a257 full, 4889d6 partial)

Source: UrsaMinor pilot matrix + escape run pack.

SCAN · TAKE IT HOME

VerdictGate

method & calculator · open source

Portfolio & contacts

victor-2026.github.io/portfolio